



EXPERIMENT: 08

Name:- Sahil

Branch:- BE-CSE

Semester:- 6th

Subject Name:- System Design

UID:- 23BCS13733

Section:- KRG1-B

Date :- 06-04-2026

Subject Code:- 23CSH-314

1.Aim:-

To design a high-level architecture for a scalable stock trading platform (similar to Zerodha, Groww, and Upstox) that supports real-time stock trading, order management, and portfolio tracking with high availability and low latency.

2.Objectives:-

The objectives of this experiment are:

1. To understand the architecture of a real-time stock trading system.
2. To design APIs for user authentication, KYC verification, market data, and order execution.
3. To analyze functional and non-functional requirements of a trading platform.
4. To study how real-time data streaming and event-driven systems work.
5. To ensure consistency in financial transactions while maintaining scalability.
6. To understand how portfolio and trade management systems operate.

3.Procedure:-

- Studied real-world stock trading platforms such as Zerodha, Groww, and Upstox to understand system behavior.
- Identified key system entities including Users, Stocks, Orders, Trades, Portfolio, and Watchlist.
- Analyzed the complete workflow from user registration → KYC → viewing stocks → placing orders → trade execution.
- Gathered functional requirements such as real-time stock updates, order placement, and portfolio tracking.
- Analyzed non-functional requirements including low latency, high availability, fault tolerance, and scalability.
- Designed RESTful APIs for user management, market data, portfolio, and watchlist features.
- Studied WebSocket-based real-time updates for stock prices.
- Understood event-driven architecture using message queues (Kafka) for trade execution and notifications.
- Evaluated trade-offs between strong consistency (orders, trades) and eventual consistency (analytics, notifications).



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

4. Functional Requirements:-

Users should be able to register, log in, and complete KYC verification.

- Users should be able to view real-time stock prices and historical data.
- The system should support placing, modifying, and canceling orders.
- Orders should provide accurate real-time status updates.
- Users should be able to maintain a watchlist with live updates.
- The system should provide a dashboard to track holdings, trade history, and P&L.
- Portfolio management should reflect executed trades instantly.

5. Non Functional Requirements:-

- The system must support millions of users simultaneously.
- Real-time stock updates should have very low latency (<100 ms).
- High availability (99.9%+) must be ensured for trading operations.
- Strong consistency is required for order execution and trades.
- The system should be fault tolerant and resilient to failures.
- Scalability should be achieved using microservices architecture.
- Secure communication and authentication (JWT, HTTPS) must be implemented.

6. Core Entities:-

- User
- Stock
- Order
- Trade
- Portfolio
- Watchlist

7. API Design:-

User APIs:

- POST /auth/signup
- POST /auth/login
- POST /auth/verify-KYC

Market Data APIs:

- WS /api/v1/stocks → Real-time stock updates
- GET /api/v1/stocks → Get all stocks
- GET /api/v1/stocks/{stock_symbol_id} → Stock details
- GET /api/v1/stocks/{stock_symbol_id}/history → Historical data

Order APIs:

- POST /api/v1/orders → Place order
- PUT /api/v1/orders/{order_id} → Modify order
- DELETE /api/v1/orders/{order_id} → Cancel order
- GET /api/v1/orders → Get user orders

Portfolio APIs:

- GET /api/v1/portfolio
- GET /api/v1/portfolio/holdings
- GET /api/v1/portfolio/positions
- GET /api/v1/portfolio/performance (Optional)

Watchlist APIs:

- GET /api/v1/watchlists
- POST /api/v1/watchlists/{watchlist_id}/symbols
- DELETE /api/v1/watchlists/{watchlist_id}/symbols/{symbol}

8.HLD:-

